



Landsnail shell $\delta^{18}\text{O}$ records in periglacial loess deposits: New transfer functions for precipitation $\delta^{18}\text{O}$ and temperature in dry continental settings

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ABSTRACT

Our knowledge about climate change is based on the availability of climate data from both the present and the past. Accurate reference data are essential for improving Earth System Models, which help us to analyse and predict climatic developments with greater accuracy. In this context, subfossil landsnail shells and their stable isotope compositions are valuable climatic proxies. They are used to derive past precipitation isotopy ($\delta^{18}\text{O}_{\text{precipitation}}$), which represents an important reference parameter for climate reconstruction. However, the interpretability of $\delta^{18}\text{O}$ shell isotopes in periglacial loess deposits has been questioned, due to the complexity of the hydrological cycle in these environments. Calibration studies investigating the relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ under continental dry conditions are still lacking. To close this gap, we collected two genera commonly found in European loess deposits (*Vallonia* and *Pupilla*) at modern dry periglacial sites in continental Inner Asia. Our results confirm a strong positive relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ of the growing season period ($\text{adj}R^2 = 0.53$, $p \ll 0.001$). When only *Vallonia* samples were considered, the relationship was even stronger ($\text{adj}R^2 = 0.60$, $p \ll 0.001$) indicating a genus-specific response. Our results support the suitability of landsnail shells as proxy for $\delta^{18}\text{O}_{\text{precipitation}}$ in dry continental settings. Furthermore, we provide an empirically derived transfer function to reconstruct $\delta^{18}\text{O}_{\text{precipitation}}$ from $\delta^{18}\text{O}_{\text{shell}}$ for periglacial loess archives as well as a transfer function for temperature approximation. This study demonstrates the high potential of shell isotopes as a quantitative climate proxy also for dry continental environments. However, future studies should focus on the resolution of shell isotope data as well as the synchronicity of calibration data, to increase data precision.

1. Introduction

Given the increasingly visible impacts of climate change on our environment, it is becoming more and more important to have reliable predictions of future developments. This also requires further improvement in the performance and accuracy of Climate and Earth System Models (see e.g., IPCC 6th Assessment Report, 2021; WMO State of the Global Climate Report, 2024). It is therefore essential to have reliable and precise reference data for calibration and cross-validation, both for the present and for the past. Data for past climates can be reconstructed from terrestrial sediments. Climatic proxies often

preserved in such deposits are gastropod shells. They have proven to be robust and valuable environmental indicators representing a well-established proxy in Quaternary science. Their fossil species compositions provide valuable information about palaeoecosystem characteristics, such as precipitation, vegetation and humidity (Ložek, 1990; Rousseau and Puisségur, 1990; Grimley et al., 2020; J. Dong et al., 2022; Sümegei et al., 2022). With the growing need for quantitative proxy data, research has increasingly focused on the isotopic composition of gastropod shells (e.g., Yanes et al., 2019; Rech et al., 2021; Richter et al., 2021; Dettman et al., 2024; Újvári et al., 2024). Shell isotopes have been used in numerous studies to infer ancient environmental and climatic

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conditions (see Prendergast et al., 2015; Colonese, 2017; Schmitt et al., 2022; Y. Dong et al., 2022; Wang et al., 2024). Calibration studies on modern terrestrial gastropods have shown, that the oxygen isotope composition of the shells ($\delta^{18}\text{O}_{\text{shell}}$) is positively correlated with the oxygen isotopic composition of precipitation ($\delta^{18}\text{O}_{\text{precipitation}}$) (e.g., Yapp, 1979; L  colle, 1985; Yanes et al., 2018; Nield et al., 2022). $\delta^{18}\text{O}_{\text{precipitation}}$ signals in turn are of great importance for palaeoclimate research. They are primarily a function of temperature (Dansgaard, 1964; Kohn and Welker, 2005; S. Wang et al., 2016), modified by site-specific elevation, rainfall intensity, continentality and the source of water vapour (e.g., Aragu  s-Aragu  s et al., 2000; Yao et al., 2013). A growing number of studies and models are attempting to disentangle the variables and complex interactions that influence $\delta^{18}\text{O}$ of palaeoprecipitation, to reconstruct precise absolute climate information (see

Vystavna et al., 2021; B  hler et al., 2022; Li et al., 2023). The prerequisite for the reconstruction of $\delta^{18}\text{O}_{\text{precipitation}}$ from fossil gastropod shells is a thorough understanding of controls on land snail body water $\delta^{18}\text{O}$ and the oxygen isotope fractionation during shell precipitation. Previous studies (e.g., Goodfriend and Magaritz, 1987; Zanchetta et al., 2005; Nield et al., 2022) have shown that $\delta^{18}\text{O}_{\text{shell}}$ primarily tracks $\delta^{18}\text{O}_{\text{precipitation}}$, while being additionally influenced by relative humidity and vapour pressure (Balakrishnan and Yapp, 2004) and the temperature during carbonate precipitation (Grossman and Ku, 1986). Furthermore, species-specific metabolic processes and activity phases influence $\delta^{18}\text{O}$ of gastropod shells (e.g., X. Wang et al., 2016), requiring consideration of the taxonomic affiliation of the analysed gastropod shell. However, it has been questioned whether the relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ is robust under arid to semi-arid

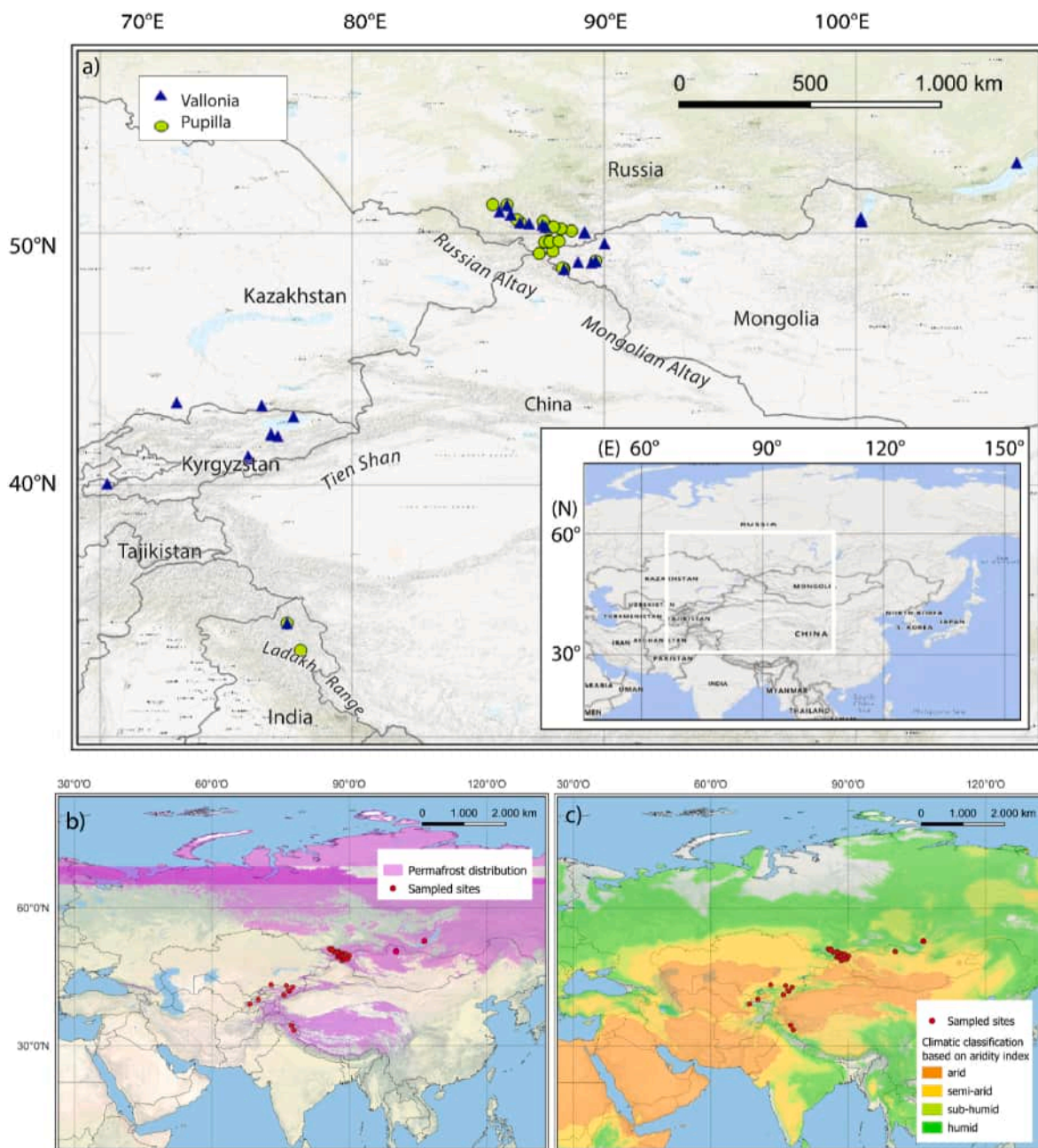


Fig. 1. Maps showing the sampled sites of this study, with a) differentiation by genus, b) depiction of the distribution of permafrost zones (Source: Biskaborn et al., 2015) and c) depiction of arid and humid climate regions across Asia for the recent period (1985–2005 CE) based on CRU (Climate Research Unit) observation data (Source: Wang and Zhang, 2020).

conditions. In dry environments, evapotranspiration significantly influences the hydrological cycle and may lead to a change in the isotopic composition of both ingested water and the snail body water from which shell carbonate is precipitated. This question is particularly important given that a large number of the available terrestrial long-term palaeoclimate archives are loess sequences (Muhs et al., 2014; Schaetzl et al., 2018; Zeeden et al., 2020; Rousseau et al., 2020). However, modern calibration studies investigating oxygen isotope incorporation into gastropod shell carbonate under dry continental conditions are still lacking. To address this, we present the first calibration study for $\delta^{18}\text{O}_{\text{shell}}$ compositions of modern landsnail shells formed under equivalent conditions, focusing on taxa commonly found in Eurasian loess deposits. The study aims to determine whether the correlation between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ remains valid under dry continental conditions analogous to those of Pleistocene loess deposition in Central Europe. Our results will help to assess the reliability of palaeoprecipitation reconstructions based on gastropod shells in loess archives and improve our understanding of past climate dynamics in periglacial environments.

2. Study area

The modern equivalent most similar to the Pleistocene loess regions of Central Europe is thought to be found in Inner Asia (Horsák et al., 2015; Chytrý et al., 2019). This area is characterised by both a dry continental climate and the presence of permafrost (see Fig. 1). The climatic similarity of these regions is further supported, for example, by the occurrence of *Vallonia tenuilabris* and *Pupilla loessica*, gastropod species that are considered as indicator species for fully glacial time conditions in Central Europe (see Ložek, 1990; Meng and Hoffmann, 2009; Horsák et al., 2015). For our calibration study, we focused on sites in continental Inner Asia with an arid to semi-arid climate combined with permafrost. Sampling sites included lowland to xeromontane ecosystems in the Mongolian and Russian Altai region, Ladakh in northern India, Kyrgyzstan, Tajikistan, northern Mongolia, and the Baikal region all of which represent dry continental cold climates. Mean annual precipitation across all study sites ranged from 100 mm to 500 mm/year with a mean of 282 mm, while mean annual temperatures ranged from 7.4 °C to −8.3 °C with a mean of −2.2 °C across all sites. The sampling locations are shown in Fig. 1 and described in more detail in the supplementary material.

3. Methods

3.1. Sampling for modern shells

Shells were collected from 65 sites across $\delta^{18}\text{O}_{\text{precipitation}}$ transects of arid to semiarid periglacial environments. These sites were distributed across four main regions comprising the Mongolian and Russian Altai (43 sites), Ladakh in northern India (3 sites), Kyrgyzstan with adjacent areas (14 sites) and northern Mongolia and Baikal area (5 sites). Within this framework, 243 gastropod shells were used, representing each site with 2–7 shells per habitat and species. We focused on the two gastropod genera *Vallonia* and *Pupilla*, as they are common taxa in European and non-monsoonal influenced Asian loess archives. While *Vallonia* and *Pupilla* do not occupy completely identical ecological niches, both are consistently associated with dry, open steppe-like habitats in our study area and show comparable environmental distributions (see also Meng and Hoffmann, 2009; Horsák et al., 2015). This ecological proximity justifies their comparison within a shared climatic framework, even though subtle differences in microhabitat preferences may exist. Due to the lack of species-specific activity data for *Vallonia* and *Pupilla* in Inner Asia, we defined the growing season as the period when mean monthly temperatures exceed 5 °C, in line with general thresholds for gastropod activity (Ansart et al., 2001; Salvador and Tomotani, 2024). However, differences in seasonal activity patterns or microhabitat preferences

between the two genera may contribute to the observed variation in shell $\delta^{18}\text{O}$ and should be kept in mind when interpreting species-specific isotope signals. To assess taxon-specific differences in shell $\delta^{18}\text{O}$ of these genera, we use shells from sites with a sympatric occurrence of *Vallonia* and *Pupilla*. Only adult shells were sampled, with shells collected predominantly empty from the soil surface. Under dry continental conditions, gastropod shells can persist for several years, due to the low rate of chemical weathering. While most shells likely represent recent growth within a few years prior to collection, a certain degree of time-averaging cannot be excluded. Consequently, $\delta^{18}\text{O}$ shell values may reflect short-term averages across a few consecutive years rather than conditions of a single year.

3.2. Stable isotope analyses

Only complete adult shells were used for isotope analyses. The shells were cleaned mechanically using an ultrasonic bath and needle and treated with hydrogen peroxide for 4 h to remove organic material. Subsequently, the shells were powdered using an agate mortar and pestle. Measurements were conducted using 0.2–0.3 mg of the homogenised carbonate powders. The carbonate powders were reacted with 100% phosphoric acid at 70 °C using a Gasbench II connected to a ThermoFisher Delta V Plus mass spectrometre. Reproducibility and accuracy were monitored by replicate analysis of laboratory standards calibrated by assigning a $\delta^{18}\text{O}$ of −2.20‰ to NBS19 and −23.2‰ to NBS18. Reproducibility for $\delta^{18}\text{O}$ analyses was ± 0.06 (1 standard deviation). International standard IAEA 603 was used as quality control sample. Throughout this article, $\delta^{18}\text{O}_{\text{shell}}$ values are reported relative to the Vienna Pee Dee Belemnite (VPDB) standard, while $\delta^{18}\text{O}_{\text{precipitation}}$ values are referenced to the Vienna Standard Mean Ocean Water (VSMOW) standard.

3.3. Data compilation with GIS and statistical analyses

To analyse the influence of environmental and climatic variables on shell isotope composition, we relied on modelled data to ensure spatial coverage for all sampling sites. All datasets used have a spatial resolution of 30 arc seconds (~925 m grid cells), and site-specific data were extracted using GIS software. Modelled $\delta^{18}\text{O}_{\text{precipitation}}$ data were provided by the International Atomic Energy Agency (IAEA) as monthly means and annual averages for the studied sites (Table 1). These data are based on the RCWIP2 model by Terzer-Wassmuth et al. (2021) (RMSE 1.29‰), which assigns each locality to one of 36 climate clusters defined by similar climate-isotope relationships. Monthly regression models are calibrated per cluster. In data-sparse regions such as Mongolia, zonal or global models are used, which may reduce predictive accuracy during specific seasons. To ensure internal consistency with the model isotope predictions, temperature values were also extracted from the RCWIP2 model outputs. These values are based on WorldClim 1.4, but may differ slightly from direct WorldClim extractions due to internal interpolation and spatial harmonisation within the RCWIP2 framework. Temperature data comprise mean annual temperature, growing season temperature (based on all month with mean temperatures >5 °C), mean temperature of the coldest month (Tmin) and the warmest month (Tmax), and the maximum annual range of monthly temperature averages. Furthermore, precipitation variables, including total amount of annual precipitation and precipitation during the warmest quarter, were obtained directly from the WorldClim database (Fick and Hijmans, 2017). All data described above refer to averaged values for the period 1970 to 2000 CE. The Global Aridity Index was obtained from Zomer et al. (2022) and is defined as the ratio between mean annual precipitation and mean annual potential evapotranspiration for the reference period of 1970–2000 CE. To assess near-surface moisture availability and potential desiccation risks for the investigated snails, we additionally used Evaporative Stress (S) extracted from the GLEAM4 dataset (Miralles et al., 2025). It is calculated as the ratio of actual evaporation (E) to

Table 1

Overview of all sampled sites in Inner Asia, including precipitation isotope data from the IAEA, $\delta^{18}\text{O}$ values calculated using Eqns (1a) and (1b) and using the flux balance model from Balakrishnan and Yapp (2004) and measured $\delta^{18}\text{O}_{\text{shell}}$ data as median values.

Sample ID	No. of anal. shells	Species	Country	Region	Latitude N	Longitude E	$\delta^{18}\text{O}_{\text{shell}}$ Median	Stddev $\delta^{18}\text{O}_{\text{shell}}$	$\delta^{18}\text{O}_{\text{prec_MA}}$	$\delta^{18}\text{O}_{\text{prec_GS}}$ > 5	$T_{\text{GS}} > 5^\circ\text{C}$	$\delta^{18}\text{O}_{\text{prec}}$ calc with Equation (1a) (this study)	$\delta^{18}\text{O}_{\text{prec}}$ calc with Equation (1b) (this study)	$\delta^{18}\text{O}_{\text{prec}}$ calc with Balakrishnan & Yapp (2004)
Mo22-076	5	<i>Vallonia tenuilabris</i>	Mongolia	Altai Region	48.86705826	89.69023082	-6.26	0.71	-19.1	-13.3	11.3	-13.8	-13.5	-16.1
Mo22-057V	3	<i>Vallonia tenuilabris</i>	Mongolia	Altai Region	48.55871688	88.42416585	-7.33	1.18	-22.2	-14.6	9.8	-14.2	-14.0	-16.7
Mo22-057P	3	<i>Pupilla loessica</i>	Mongolia	Altai Region	48.55871688	88.42416585	-4.81	0.80	-22.2	-14.6	9.8	-13.1	-12.9	-15.0
Mo22-055	5	<i>Pupilla loessica</i>	Mongolia	Altai Region	48.55910438	88.42436931	-4.74	0.60	-22.2	-14.6	9.8	-13.1	-12.9	-14.8
Mo22-049	3	<i>Pupilla loessica</i>	Mongolia	Altai Region	48.59461056	88.35506513	-5.35	0.59	-21.6	-13.5	7.9	-13.4	-13.1	-15.5
Mo22-035	5	<i>Vallonia tenuilabris</i>	Mongolia	Altai Region	48.83347749	88.97933911	-6.08	0.76	-19.7	-13.6	9.9	-13.7	-13.4	-15.8
Mo22-022	5	<i>Vallonia tenuilabris</i>	Mongolia	Altai Region	49.57681443	90.03213951	-5.91	0.71	-18.9	-13.1	9.6	-13.6	-13.4	-15.6
Mo22-007V	3	<i>Vallonia tenuilabris</i>	Mongolia	Altai Region	48.87402819	89.67917583	-4.58	1.35	-19.1	-13.3	11.4	-13.1	-12.8	-15.2
Mo22-007P	2	<i>Pupilla loessica</i>	Mongolia	Altai Region	48.87402819	89.67917583	-4.31	0.08	-19.1	-13.3	11.4	-12.9	-12.7	-14.3
Mo22-078	3	<i>Vallonia tenuilabris</i>	Mongolia	Altai Region	48.80773227	89.51758904	-6.65	0.36	-18.7	-13.1	11.3	-13.9	-13.7	-16.3
4014	3	<i>Vallonia tenuilabris</i>	Mongolia	Northern Mongolia	50.46336111	100.2471667	-6.11	0.86	-19.9	-13.7	9.7	-13.7	-13.5	-16.0
3970	3	<i>Vallonia kamtschatica</i>	Mongolia	Northern Mongolia	50.46441667	100.24325	-5.18	2.14	-19.9	-13.7	9.7	-13.3	-13.1	-15.7
3969	3	<i>Vallonia tenuilabris</i>	Mongolia	Northern Mongolia	50.60047222	100.207	-5.67	0.13	-18.5	-13.0	9.5	-13.5	-13.3	-15.8
3960	3	<i>Vallonia tenuilabris</i>	Mongolia	Northern Mongolia	50.45652778	100.1853889	-6.34	0.19	-19.6	-13.6	10.3	-13.8	-13.6	-16.3
VPL219	5	<i>Vallonia costata</i>	Russia	Altai Region	50.40611111	86.67666667	-4.89	0.81	-16.5	-12.2	14.2	-13.2	-12.9	-14.8
VPH130	4	<i>Vallonia costata</i>	Russia	Altai Region	51.10472222	86.16027778	-3.38	1.11	-15.3	-12.1	13.1	-12.5	-12.3	-12.9
PPH130	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	51.10472222	86.16027778	-4.67	1.20	-15.3	-12.1	13.1	-13.1	-12.9	-13.8
PMC102	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.24888889	87.69222222	-2.58	1.10	-18.9	-13.4	10.6	-12.2	-12.0	-12.7
PL445	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.23333333	87.78916667	-4.63	1.03	-19.2	-13.7	11.4	-13.1	-12.8	-14.2
PL440	5	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.07277778	88.72	-4.07	0.72	-20.6	-14.1	9.8	-12.8	-12.6	-13.8
PL423	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	49.26861111	87.98527778	-4.82	1.21	-22.0	-13.7	6.6	-13.2	-12.9	-15.7
PL219	4	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.40611111	86.67666667	-4.06	0.83	-16.5	-12.2	14.2	-12.8	-12.6	-13.5
PH128	2	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.30805556	87.65194444	-4.74	0.53	-18.0	-12.8	10.9	-13.1	-12.9	-15.0
PH127	4	<i>Pupilla alpicola</i>	Russia	Altai Region	50.30666667	87.64861111	-3.55	1.83	-18.0	-12.8	10.8	-12.6	-12.4	-14.5
PH119	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.16416667	87.80861111	-4.14	0.73	-19.9	-14.2	9.4	-12.9	-12.6	-14.1

(continued on next page)

Table 1 (continued)

Sample ID	No. of anal. shells	Species	Country	Region	Latitude N	Longitude E	$\delta^{18}\text{O}_{\text{shell}}$ Median	Stddev $\delta^{18}\text{O}_{\text{shell}}$	$\delta^{18}\text{O}_{\text{prec_MA}}$	$\delta^{18}\text{O}_{\text{prec_GS}}$ > 5	T _{GS} >5°C	$\delta^{18}\text{O}_{\text{prec}}$ calc with Equation (1a) (this study)	$\delta^{18}\text{O}_{\text{prec}}$ calc with Equation (1b) (this study)	$\delta^{18}\text{O}_{\text{prec}}$ calc with Balakrishnan & Yapp (2004)
PH118	2	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.14861111	87.81583333	-4.62	0.68	-19.9	-13.3	10.4	-13.1	-12.8	-14.8
PH117	3	<i>Pupilla alpicola</i>	Russia	Altai Region	50.15111111	88.30277778	-6.04	0.29	-19.4	-12.9	11.3	-13.7	-13.4	-16.0
MH4d	3	<i>Pupilla turcmenia</i>	Russia	Altai Region	49.175022	87.440864	-4.90	0.11	-21.7	-13.2	7.0	-13.2	-13.0	-15.7
MC436	4	<i>Pupilla alpicola</i>	Russia	Altai Region	51.11444444	85.59638889	-3.46	0.49	-16.3	-11.6	10.6	-12.6	-12.3	-13.5
MC429	3	<i>Pupilla alpicola</i>	Russia	Altai Region	49.62861111	87.65666667	-3.37	0.37	-20.4	-13.1	8.1	-12.5	-12.3	-14.0
MC409	6	<i>Pupilla alpicola</i>	Russia	Altai Region	49.635	87.86194444	-3.86	0.54	-20.2	-13.1	9.5	-12.7	-12.5	-14.1
MC408	4	<i>Pupilla turcmenia</i>	Russia	Altai Region	49.66083333	88.2275	-2.96	1.06	-21.7	-13.6	6.7	-12.4	-12.1	-13.8
MC221	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.51583333	86.55166667	-3.47	0.45	-16.3	-12.1	14.9	-12.6	-12.3	-12.7
MC218	4	<i>Vallonia costata</i>	Russia	Altai Region	50.36916667	87.04	-5.33	1.09	-16.3	-11.8	12.7	-13.4	-13.1	-15.9
MC216	6	<i>Pupilla turcmenia</i>	Russia	Altai Region	50.23138889	88.00916667	-5.30	1.56	-19.8	-14.2	12.2	-13.4	-13.1	-14.5
MC209	6	<i>Pupilla alpicola</i>	Russia	Altai Region	50.44722222	87.60777778	-6.09	0.41	-19.9	-13.1	8.4	-13.7	-13.4	-16.2
MC102	6	<i>Vallonia costata</i>	Russia	Altai Region	50.24888889	87.69222222	-3.21	1.12	-18.9	-13.4	10.6	-12.5	-12.2	-13.6
JD223	4	<i>Vallonia costata</i>	Russia	Altai Region	50.84638889	85.86305556	-5.09	1.33	-16.7	-12.1	12.7	-13.3	-13.0	-14.8
4551	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.0167	89.24003333	-4.82	0.27	-21.5	-13.9	8.4	-13.2	-12.9	-14.7
4549	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.01561667	89.24638333	-6.34	0.43	-21.7	-13.8	7.4	-13.8	-13.6	-16.1
4545	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.01906667	89.23308333	-4.97	0.61	-21.4	-13.9	9.2	-13.2	-13.0	-14.6
4544	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.02056667	89.24303333	-5.58	1.28	-21.8	-13.9	7.4	-13.5	-13.2	-15.7
4542	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.02091667	89.24015	-6.17	0.77	-21.5	-13.9	8.4	-13.7	-13.5	-16.1
4540	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.00676667	89.24371667	-5.77	0.41	-22.1	-13.9	5.9	-13.6	-13.3	-15.5
4532	3	<i>Vallonia tenuilabris</i>	Russia	Altai Region	50.31588333	87.56521667	-5.98	0.81	-17.8	-12.7	11.0	-13.7	-13.4	-16.2
4627	3	<i>Vallonia kamtschatica</i>	Russia	Altai Region	50.72591667	86.30938333	-3.71	0.55	-16.4	-12.1	13.9	-12.7	-12.4	-13.4
4622	3	<i>Vallonia costata</i>	Russia	Altai Region	50.72746667	86.3107	-5.23	1.00	-16.4	-12.1	13.9	-13.3	-13.1	-14.5
2524	3	<i>Vallonia kamtschatica</i>	Russia	Baikal region	52.78333333	106.4	-7.62	0.11	-18.9	-13.5	11.3	-14.4	-14.1	-17.3
5341	7	<i>Pupilla turcmenia</i>	India	Ladakh Region	33.40927778	77.96236111	-4.77	1.06	-19.1	-14.5	7.4	-13.1	-12.9	-15.5
5336	4	<i>Pupilla turcmenia</i>	India	Ladakh Region	34.48833333	77.432528	-4.85	1.21	-17.2	-14.2	8.7	-13.2	-12.9	-15.6
5335	3	<i>Vallonia ladacensis</i>	India	Ladakh Region	34.48833333	77.432528	-4.47	0.73	-17.2	-14.2	8.7	-13.0	-12.8	-15.2

(continued on next page)

Table 1 (continued)

Sample ID	No. of anal. shells	Species	Country	Region	Latitude N	Longitude E	$\delta^{18}\text{O}_{\text{shell}}$ Median	Stddev $\delta^{18}\text{O}_{\text{shell}}$	$\delta^{18}\text{O}_{\text{prec_MA}}$	$\delta^{18}\text{O}_{\text{prec_GS}}$ > 5	$T_{\text{GS}>5^\circ\text{C}}$	$\delta^{18}\text{O}_{\text{prec}}$ calc with Equation (1a) (this study)	$\delta^{18}\text{O}_{\text{prec}}$ calc with Equation (1b) (this study)	$\delta^{18}\text{O}_{\text{prec}}$ calc with Balakrishnan & Yapp (2004)
5378	3	<i>Vallonia costata</i>	Kazakhstan	Kyrg. & adjacent areas	43.13858333	76.38611111	−0.93	0.42	−13.2	−10.6	15.1	−11.5	−11.3	−9.9
2911	3	<i>Vallonia tenuilabris</i>	Kyrgyzstan	Kyrg. & adjacent areas	43.25925	73.00486111	−0.87	0.42	−11.5	−9.2	19.0	−11.5	−11.3	−9.0
4876	3	<i>Vallonia pulchella</i>	Kyrgyzstan	Kyrg. & adjacent areas	42.70569444	77.68569444	−3.00	0.75	−12.7	−10.5	12.6	−12.4	−12.1	−12.2
1972	2	<i>Vallonia asiatica</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.91666667	77.06666667	−2.07	0.33	−20.6	−11.5	5.2	−12.0	−11.8	−12.9
4870	3	<i>Vallonia tenuilabris</i>	Kyrgyzstan	Kyrg. & adjacent areas	42.03847222	76.77647222	−0.21	0.29	−16.7	−11.7	9.0	−11.2	−11.0	−10.4
4868	3	<i>Vallonia tenuilabris</i>	Kyrgyzstan	Kyrg. & adjacent areas	42.00677778	76.76325	−0.04	0.78	−18.0	−11.8	8.4	−11.1	−10.9	−10.1
4866	3	<i>Vallonia tenuilabris</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.97180556	76.77005556	−0.08	0.25	−19.8	−11.6	6.3	−11.1	−10.9	−10.3
3342	3	<i>Vallonia tenuilabris</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.04794444	75.88536111	0.02	1.13	−19.8	−10.8	6.2	−11.1	−10.9	−10.4
4864	3	<i>Vallonia costata</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.09852778	75.82727778	1.23	1.83	−16.7	−11.6	12.4	−10.6	−10.4	−8.5
4861	3	<i>Vallonia costata</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.09397222	75.84252778	0.78	0.24	−16.9	−11.8	11.7	−10.7	−10.6	−8.7
4860	3	<i>Vallonia tenuilabris</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.04136111	75.87652778	1.04	0.11	−19.3	−10.6	7.1	−10.6	−10.5	−9.8
4862	3	<i>Vallonia asiatica</i>	Kyrgyzstan	Kyrg. & adjacent areas	41.09397222	75.84252778	−0.99	0.33	−16.9	−11.8	11.7	−11.5	−11.3	−10.4
4126	3	<i>Vallonia asiatica</i>	Kyrgyzstan	Kyrg. & adjacent areas	40.03366667	70.24865	1.22	1.13	−10.3	−8.9	15.7	−10.6	−10.4	−8.1
5033	1	<i>Vallonia asiatica</i>	Tajikistan	Kyrg. & adjacent areas	39.05138889	68.39122222	0.39		−16.7	−9.4	13.1	−10.9	−10.7	−8.9

potential evaporation (Ep), integrating information from process-based modelling and satellite data. The dataset incorporates factors such as solar radiation, wind speed, vapour pressure deficit, surface and root-zone soil moisture and vegetation characteristics, thereby approximating the moisture conditions experienced by small ground-dwelling organisms. GLEAM4 data are available at a spatial resolution of 0.1° (~ 10 km) and were extracted as annual mean values for the period 2000 to 2022 CE.

We applied a Shapiro-Wilk test to assess the normality of the data distributions, with a p-value < 0.05 and a W-value close to 1 indicating a normal distribution. The Spearman correlation test was used to compare pairwise relationships among all variables. This method uses ranked data and takes into account potential non-linear relationships, making it robust to non-normally distributed data. To develop a predictive model for $\delta^{18}\text{O}_{\text{precipitation}}$ based on $\delta^{18}\text{O}_{\text{shell}}$, we performed multiple linear regression with manual variable selection, to determine the best-fitting

model. This manual approach ensures that only the most relevant predictors are retained while avoiding overfitting due to multicollinearity. In addition, we used least absolute shrinkage and selection operator (LASSO) regression, which is a regularisation method for linear regression models. It reduces errors caused by overfitting and collinearity by using subsets of all samples as training data. To examine taxon-related influences on shell isotopes and to correct for spatial autocorrelation, generalised estimating equations (GEE; `geeglm` function from the `geepack` R package by Højsgaard et al., 2006) with a Gaussian error structure were used to test differences in isotope signals between snail taxa using R software 4.1.0 (R Core Team, 2021).

4. Results

The overall range for $\delta^{18}\text{O}_{\text{precipitation}}$ across all sites was 5.65‰ (−8.9‰ to −14.6‰) for the growing season and 11.9‰ (−10.3‰ to

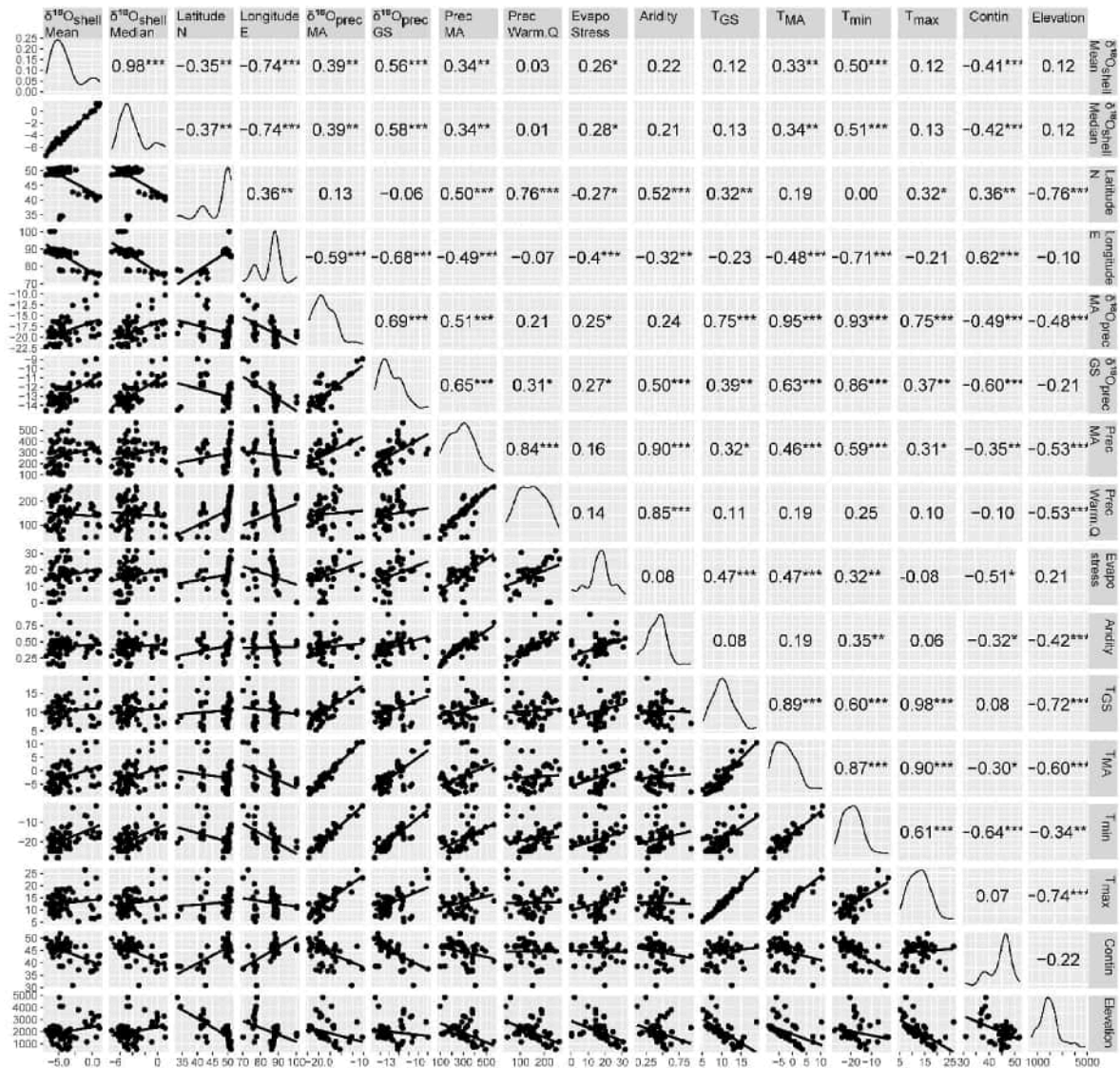


Fig. 2. The figure depicts pairwise relationships between the investigated climatic and spatial variables and stable oxygen isotope values ($\delta^{18}\text{O}$) in shells and precipitation. The matrix is based on Spearman correlation coefficients alongside scatter plots for visualising associations between variables: $\delta^{18}\text{O}_{\text{shell}}$ mean; $\delta^{18}\text{O}_{\text{shell}}$ median; Latitude ($^\circ\text{N}$); Longitude ($^\circ\text{E}$); $\delta^{18}\text{O}_{\text{prec}}$ MA, mean annual precipitation $\delta^{18}\text{O}$ (‰ SMOW); $\delta^{18}\text{O}_{\text{prec}}$ GS, growing season precipitation $\delta^{18}\text{O}$ (‰ SMOW) for months with average mean temperatures of $> 5^\circ\text{C}$; Prec MA, mean annual precipitation amount (mm); PrecWarmQ, warmest quarter precipitation amount (mm); Evapo stress, mean annual Evaporative Stress after Miralles et al. (2025); Aridity, Global Aridity Index after Zomer et al. (2022); TGS, growing season temperature ($^\circ\text{C}$); TMA, mean annual temperature ($^\circ\text{C}$); Tmin, minimum, annual temperature ($^\circ\text{C}$); Tmax, maximum annual temperature ($^\circ\text{C}$); Contin, continentality ($\Delta^\circ\text{C}$); Elevation (m asl). Significance: ***, < 0.001 ; **, < 0.01 ; *, < 0.05 ; not shown, > 0.05 . Diagonal cells, where identical variables intersect, display the respective frequency distributions.

–22.2‰) in annual average. For specific subregions, $\delta^{18}\text{O}_{\text{precipitation}}$ ranged from –14.6‰ to –11.6‰ for the Altai (total range of 3.0‰), from –13.71‰ to –13.03‰ for Northern Mongolia and the Baikal area (total range of 0.68‰), –14.5‰ to –14.2‰ for Ladakh (total range of 0.31‰) and from –11.8‰ to –8.9‰ for Kyrgyzstan and adjacent areas (total range of 2.9‰).

Compared to $\delta^{18}\text{O}_{\text{precipitation}}$, median $\delta^{18}\text{O}_{\text{shell}}$ values across all sites ranged from –7.6‰ to 1.2‰. Their total range within the different subregions was 4.8‰ for the Altai, 2.4‰ for Northern Mongolia and the Baikal area, 0.4‰ for Ladakh and 4.2‰ for Kyrgyzstan and adjacent areas. The standard deviations for the mean ranged from $\pm 0.08\text{‰}$ to $\pm 2.86\text{‰}$. To reduce noise in the dataset, sites with a standard deviation above the 95th percentile (2.2‰) were excluded from further analysis. The mean standard deviation of the remaining samples was $\pm 0.76\text{‰}$. Site-specific results of the $\delta^{18}\text{O}_{\text{shell}}$ measurements are presented in Table 1, while the complete dataset including individual shell measurements is provided in the supplementary file. $\delta^{18}\text{O}_{\text{shell}}$ values showed a strong positive correlation with $\delta^{18}\text{O}_{\text{precipitation}}$ of the growing season ($r_s = 0.58$) as shown in Figs. 2 and 3, and a moderate correlation with mean annual $\delta^{18}\text{O}_{\text{precipitation}}$ ($r_s = 0.39$). Moreover, we found $\delta^{18}\text{O}_{\text{shell}}$ to strongly correlate with longitude ($r_s = -0.74$) and minimum temperature ($r_s = 0.51$). Mean annual temperature moderately correlated with $\delta^{18}\text{O}_{\text{shell}}$ ($r_s = 0.34$), while very low correlation was observed between shell signals and growing season temperature ($r_s = 0.13$). While annual precipitation amount moderately correlated with shell isotopy, ($r_s = 0.33$), there was no correlation with precipitation amount of the warmest quarter ($r_s = 0.01$). Furthermore, weak correlation was found with Aridity Index ($r_s = 0.21$) and Evaporative Stress ($r_s = 0.28$), suggesting that although evaporative conditions may influence the isotopic baseline, they do not primarily control $\delta^{18}\text{O}_{\text{shell}}$ values, which appear to be predominantly governed by precipitation isotopy. Fig. 2 shows a correlation panel of the relevant relationships.

$\delta^{18}\text{O}_{\text{shell}}$ values for the genus *Vallonia* were on average more negative (mean = –5.1‰) compared to those of *Pupilla* (mean = –4.3‰) if compared over shared sites. However, GEE model based on all individual measurements ($n = 56$), accounting for spatial allocation indicated that this difference was not statistically significant ($X^2(1) = 0.19$, $p = 0.66$). When analysing the relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ for each of those taxa, we found a strong positive linear relationship for samples allocated to *Vallonia* ($\text{adj}R^2 = 0.60$, $p < 0.001$), but interestingly no statistically significant relationship for *Pupilla* ($\text{adj}R^2 = 0.02$, $p = 0.242$; see Fig. 3).

To assess the quantitative relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ in dry periglacial environments, we developed a

regression model based on our empirical data from Inner Asian periglacial sites. As the previous results indicated a stronger correlation when using exclusively shells of the genus *Vallonia*, we provide equations for the complete dataset:

$$\delta^{18}\text{O}_{\text{precipitation (growing season)}} = -11.08 + 0.43 \times \delta^{18}\text{O}_{\text{shell}} \quad (r = 0.73, r^2 = 0.54, \text{RMSE} = 0.92\text{‰}) \quad (1a)$$

and exclusively for data from *Vallonia*:

$$\delta^{18}\text{O}_{\text{precipitation (growing season)}} = -10.89 + 0.42 \times \delta^{18}\text{O}_{\text{shell}} \quad (r = 0.8, r^2 = 0.64, \text{RMSE} = 0.89\text{‰}) \quad (1b)$$

Fig. 4 depicts the $\delta^{18}\text{O}_{\text{precipitation}}$ values calculated using Eqns (1a) and (1b) as well as the values obtained using the mechanistic model of Balakrishnan and Yapp (2004), plotted against the modern $\delta^{18}\text{O}_{\text{precipitation}}$ reference data from the IAEA.

To assess the robustness of Eqns (1a) and (1b), we tested the influence of additional variables on $\delta^{18}\text{O}_{\text{shell}}$ using LASSO regression, which reduces collinearity and overfitting by selecting only the most predictive variables. Using an interaction level of 1, which considers also pairwise interactions between the variables, and setting λ to 0.1, the strongest predictor of $\delta^{18}\text{O}_{\text{shell}}$ beyond $\delta^{18}\text{O}_{\text{precipitation}}$ of the growing season ($\beta = 0.710$) was growing season temperature ($\beta = -0.153$), followed by continentality ($\beta = -0.105$), aridity ($\beta = -0.056$), mean annual precipitation ($\beta = -0.025$) and evapotranspiration ($\beta = 0.019$). Longitude, latitude and elevation were excluded from testing as they combine the effects of various climatic and local variables.

5. Discussion

5.1. $\delta^{18}\text{O}_{\text{shell}}$ and its relationship to other variables

The $\delta^{18}\text{O}_{\text{shell}}$ data from dry periglacial sites in Inner Asia revealed a significant positive correlation between $\delta^{18}\text{O}_{\text{shell}}$ and annual average $\delta^{18}\text{O}_{\text{precipitation}}$ and an even stronger correlation with growing season $\delta^{18}\text{O}_{\text{precipitation}}$. This indicates that the growing season period more accurately represents the period recorded in the shell, relating to the snail activity phase and time of shell growth. However, the activity phases of micro-snails in periglacial regions are not well studied, yet. Previous studies suggest that some snails have a supercooling ability and can be active even around 0 °C (Ansart et al., 2001; Salvador and Tomotani, 2024). Based on our results, we assume an activity phase corresponding to mean monthly temperatures above 5 °C, as related $\delta^{18}\text{O}_{\text{precipitation}}$ values showed the strongest correlation with shell

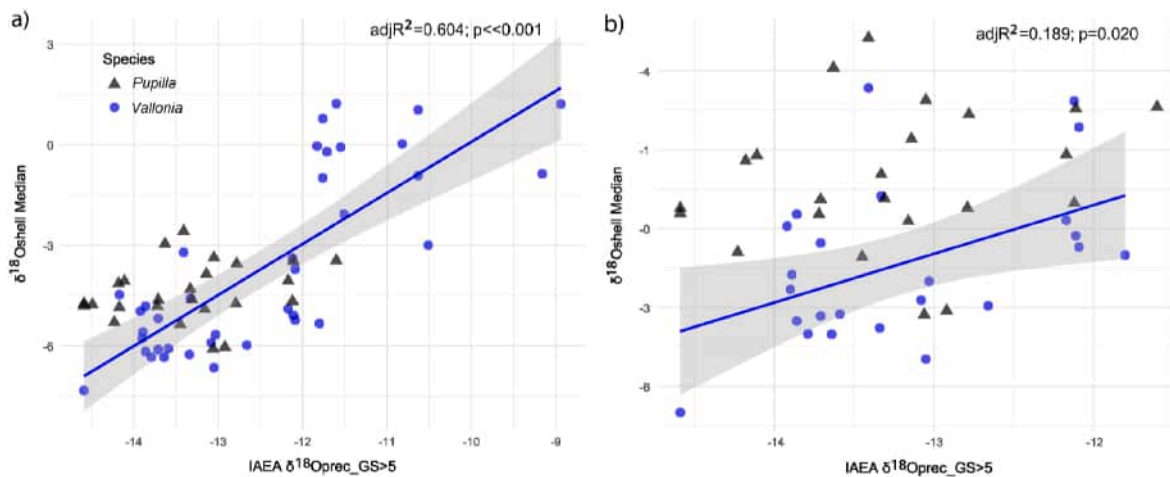


Fig. 3. Relationship between shell and growing season precipitation $\delta^{18}\text{O}$ values. Isotope data for *Pupilla* and *Vallonia* shells were analysed separately, but no significant relationship was detected for *Pupilla*: (a) the complete dataset and (b) the Altai subregion alone. Adjusted R^2 values from linear regression and their significance shown in the figures refer to *Vallonia* data.

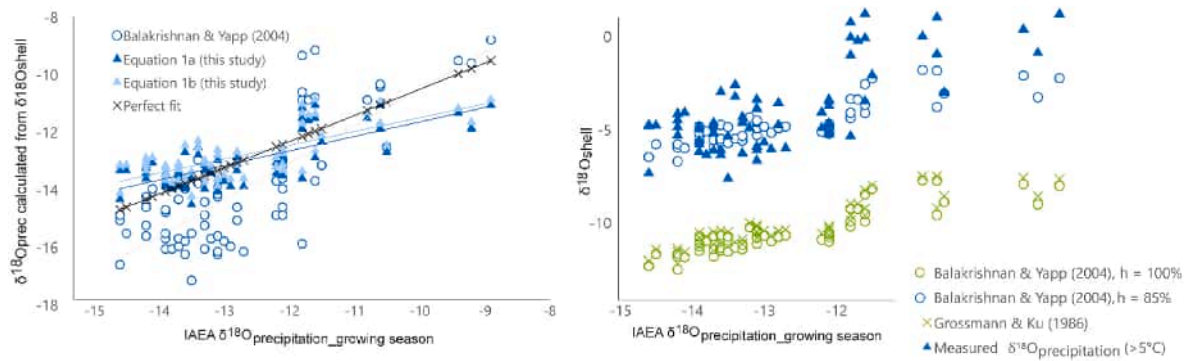


Fig. 4. Left: Relationship between modern growing season $\delta^{18}\text{O}_{\text{precipitation}}$ data from IAEA and growing season $\delta^{18}\text{O}_{\text{precipitation}}$ data reconstructed from snail shell carbonate. For the latter we used three different formulas that are I) the empirical regression equations of this study: Eqn. (1a) (dark blue triangles) and Eqn. (1b) (light blue triangles) and II) the mechanistic model by Balakrishnan and Yapp (2004) (open black circles). The 1:1 line (black crosses) represents perfect agreement between reconstructed and observed precipitation $\delta^{18}\text{O}$. Right: Relationship between modern growing season $\delta^{18}\text{O}$ values of precipitation and $\delta^{18}\text{O}$ values of landsnail shells. Measured $\delta^{18}\text{O}_{\text{shell}}$ values from our study area (blue triangles) are compared to theoretical $\delta^{18}\text{O}_{\text{shell}}$ values calculated from growing-season precipitation $\delta^{18}\text{O}$ and temperature under different model assumptions: (a) Balakrishnan and Yapp (2004) without evaporation ($h = 1.0$; open green circles), (b) Balakrishnan and Yapp (2004) with evaporation ($h = 0.85$; open blue circles), and (c) Grossman and Ku (1986) without evaporation (green crosses). The comparison shows that only the model incorporating evaporative enrichment reproduces the measured shell $\delta^{18}\text{O}$ values, while models assuming isotopic equilibrium systematically underestimate $\delta^{18}\text{O}$, indicating a significant effect of evaporation on shell isotopic composition in arid environments.

isotopes, when compared to $\delta^{18}\text{O}$ for mean monthly averages above 0°C and 10°C . The correlation between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ aligns with the results of previous studies for humid regions (see L  colle, 1985; Zanchetta et al., 2005; Nield et al., 2022), indicating that this relationship also holds under arid to semiarid continental conditions. In this study area, $\delta^{18}\text{O}_{\text{shell}}$ values are on average -4.0‰ , corresponding to an enrichment of 8.8‰ compared to mean growing season $\delta^{18}\text{O}_{\text{precipitation}}$ (-12.8‰) and 14.6‰ relative to annual $\delta^{18}\text{O}_{\text{precipitation}}$ (-18.6‰). This enrichment exceeds the values reported for humid regions, where shells show smaller offsets from precipitation $\delta^{18}\text{O}$. For instance, in Alaska, $\delta^{18}\text{O}_{\text{shell}}$ is enriched by 5.7‰ relative to the growing season and 8.9‰ relative to annual precipitation. In Scandinavia, the enrichment is 5.6‰ and 8.0‰ , respectively (Nield et al., 2022), and in Central Europe, the enrichment is 6.3‰ relative to annual $\delta^{18}\text{O}_{\text{precipitation}}$ (L  colle, 1985). The comparatively higher enrichment of ^{18}O in the shells from our arid to subarid study area likely reflects stronger evaporation, affecting the ingested environmental water and the snail body fluid from which the shell is precipitated. Fig. 4 shows measured $\delta^{18}\text{O}_{\text{shell}}$ values from our study sites compared to theoretical $\delta^{18}\text{O}_{\text{shell}}$ values calculated from growing-season precipitation $\delta^{18}\text{O}$ and temperature under different model assumptions. The comparison shows that only the model incorporating evaporation reproduces the measured shell $\delta^{18}\text{O}$ values, while models assuming isotopic equilibrium systematically underestimate $\delta^{18}\text{O}$, indicating that all shell $\delta^{18}\text{O}$ values in our dataset underlie evaporative enrichment. Despite that, within our dry continental study area, evaporation appears to exert only a subordinate control on shell $\delta^{18}\text{O}$ variation, as suggested by the lack of correlation with either evaporative stress or aridity index. This limited influence could be due to the relatively small aridity gradient within the study area, reducing the variability threshold for detecting a clear evapotranspiration effect. Alternatively, taxon-specific differences may play a role, as the species investigated here differ from those in the other studies. In contrast to evapotranspiration and aridity, which seem to have little influence in our dataset, $\delta^{18}\text{O}_{\text{shell}}$ values show a strong correlation with both, longitude and latitude. This pattern was also reported by Nield et al. (2022) for higher latitudes in Europe and North America. However, longitude and latitude do not function as direct drivers for variation in shell $\delta^{18}\text{O}$, they rather represent environmental gradients likely reflecting collinearity effects with the most strongly correlated factors

such as $\delta^{18}\text{O}_{\text{precipitation}}$ and minimum temperature. However, geographic coordinates can also capture additional influences, such as photoperiod length varying with latitude, which influences seasonal activity patterns (Salvador and Tomotani, 2024). These activity patterns in turn influence the span of $\delta^{18}\text{O}_{\text{precipitation}}$ signals incorporated in the snail shells. Similarly, increasing continentality with longitude and latitude can shorten the duration of snail activity phases due to more extreme temperature fluctuations and extensive periods of unfavourable conditions. This could also explain the observed correlation between $\delta^{18}\text{O}_{\text{shell}}$ and minimum temperature, even though snails are generally inactive during the coldest seasons. In addition, data structure effects may have influenced the observed correlation with geographical coordinates. A more clustered distribution of Kyrgyzstan samples with a lower variability in $\delta^{18}\text{O}_{\text{shell}}$ could have strengthened the regression. This may mathematically increase the correlation with longitude, as differences between regions appear larger. Based on these considerations, we focus on $\delta^{18}\text{O}_{\text{precipitation}}$ as the primary driver of shell isotope composition, as it represents the most direct and significant environmental influence on $\delta^{18}\text{O}_{\text{shell}}$ and provides a robust basis for further interpretation.

5.2. Taxon-related differences in $\delta^{18}\text{O}_{\text{shell}}$

Previous studies have shown (e.g., Yanes et al., 2019) that shell $\delta^{18}\text{O}$ signals can vary between species, due to differences in activity phases, metabolism and feeding behaviour. The heterogeneous distribution of species through the study area impeded a direct comparison of $\delta^{18}\text{O}_{\text{shell}}$ signals of the same species across all locations. In order to identify taxon-specific differences in $\delta^{18}\text{O}_{\text{shell}}$, we analysed $\delta^{18}\text{O}_{\text{shell}}$ values for two genera, *Vallonia* and *Pupilla*, each comprising multiple species. $\delta^{18}\text{O}_{\text{shell}}$ signals showed no statistically significant difference between *Vallonia* and *Pupilla* at shared locations. This suggests that site-specific environmental variables like $\delta^{18}\text{O}_{\text{precipitation}}$ had a stronger influence on $\delta^{18}\text{O}_{\text{shell}}$ than taxon-specific aspects, which aligns with observations from previous authors (see Yanes et al., 2019). However, when examining taxon-specific correlations between shell $\delta^{18}\text{O}$ and $\delta^{18}\text{O}_{\text{precipitation}}$, we found a higher positive relationship when using only *Vallonia* data ($\text{adj}R^2 = 0.60$, $p < 0.001$), but interestingly, correlations for *Pupilla* data were very weak and not statistically significant. One possible explanation for this discrepancy is that *Pupilla* may reflect stronger fluctuations

in $\delta^{18}\text{O}_{\text{precipitation}}$. As shell growth for *Pupilla* usually exceeds one vegetation period, individual shells may represent different seasonal windows (e.g., with one individual including spring signals of the following year, while another one does not), leading to additional variability in $\delta^{18}\text{O}_{\text{shell}}$ values. This variability could weaken the correlation with $\delta^{18}\text{O}_{\text{precipitation}}$, even though *Pupilla* still incorporates precipitation signals. Moreover, sampling sites for *Pupilla* covered a smaller total range in both $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ (3.5‰ and 3.0‰) compared to the sampling area of *Vallonia* (8.8‰ and 5.7‰), which would weaken this relationship as well. With the current resolution, *Vallonia* seems to track

water ingested by snails. The model is based on the theoretical relationships between $\delta^{18}\text{O}_{\text{shell}}$ and the variables that directly influence it as a function of relative humidity, temperature, ingested water and input-output flux ratios of H_2O into the body fluid. The $\delta^{18}\text{O}$ of the shell carbonate is calculated as:

$$\delta^{18}\text{O}_{\text{shell}} = \delta^{18}\text{O}_{\text{body fluid}} + \frac{19.7 - T}{4.34} \quad (2)$$

(Balakrishnan and Yapp, 2004, modified from Grossman and Ku, 1986; T = temperature in °C). The $\delta^{18}\text{O}$ of the snail body fluid is calibrated as:

$$\delta^{18}\text{O}_{\text{body fluid}} = \frac{{}^{18}\text{a}_k \cdot {}^{18}\text{a}_{w-v} (1 - h) \cdot (\delta^{18}\text{O}_{\text{IN}} + 1000) + h \cdot (\delta^{18}\text{O}_{\text{IN}} + 1000) \cdot (1 - \theta)}{(1 - \theta) + {}^{18}\text{a}_k \cdot {}^{18}\text{a}_{w-v} (1 - h) \cdot \theta} - 1000 \quad (3)$$

$\delta^{18}\text{O}_{\text{precipitation}}$ more accurately, representing a more reliable basis for palaeoclimatic reconstructions. Since *Pupilla* is more frequently found in loess deposits, future high-resolution studies using sclerochronological analyses may help to refine its applicability for palaeoclimate reconstructions.

5.3. Comparison of different regions

The dataset reflects a spatial pattern in $\delta^{18}\text{O}_{\text{shell}}$ which corresponds to variations in climate variables and regional $\delta^{18}\text{O}_{\text{precipitation}}$. To test whether this pattern truly reflects $\delta^{18}\text{O}_{\text{precipitation}}$ isotopy or is instead shaped by any confounding unmeasured variable, we also examined $\delta^{18}\text{O}_{\text{shell}}$ relationships within individual subregions.

However, when analysing smaller spatial subsets, the correlation between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ was not statistically significant in the majority of subregions due to the narrower range in $\delta^{18}\text{O}_{\text{precipitation}}$ signals. Lower variability in $\delta^{18}\text{O}_{\text{precipitation}}$ in subregions makes trends less visible, as random fluctuations such as analytical errors or local influences carry more weight relative to the overall variation. Consequently, underlying relationships may appear weaker or become statistically insignificant. To assess, whether the lack of correlation within subregions is representative or biased by statistical noise, we evaluated the data reliability using a common significance threshold for normally distributed data (Saphiro-Wilk test: W = 0.94, p = 0.002) of six times the standard deviation (=4.6‰) comparing it to the range of $\delta^{18}\text{O}_{\text{shell}}$ values. Within subregions, $\delta^{18}\text{O}_{\text{shell}}$ variability ranged from 0.4‰ to 4.4‰ remaining below this threshold. This suggests that the variability within subregions may not sufficiently capture broader trends and is more likely caused by local influences and statistical noise. Even though all subregions exhibit lower isotopic variability than this threshold, we focus on the Altai subregion with the largest internal range to test whether a correlation between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ is detectable if applying a significance threshold of p ≤ 0.05. When analysing data for all species combined, the relationship was very weak and insignificant ($\text{adjR}^2 = 0.06$, p = 0.058). However, for exclusively *Vallonia*, which tracks $\delta^{18}\text{O}_{\text{precipitation}}$ more reliably, the analysis replicated the trend observed in the complete dataset, yielding a weaker but still significant correlation with $\delta^{18}\text{O}_{\text{precipitation}}$ ($\text{adjR}^2 = 0.19$, p = 0.020). This suggests that while local-scale variability reduces correlation strength, the broader relationship remains detectable when taxonomic consistency is ensured, supporting the robustness of the model.

5.4. Quantitative reconstruction of $\delta^{18}\text{O}_{\text{precipitation}}$ from $\delta^{18}\text{O}_{\text{shell}}$ for dry periglacial areas

Balakrishnan and Yapp (2004) developed a mechanistic flux balance model for the quantitative relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}$ of the

(Balakrishnan and Yapp, 2004; h = relative humidity, ${}^{18}\text{a}_k = 1.0284$, ${}^{18}\text{a}_{w-v}$ = equilibrium oxygen isotopic fractionation factor between liquid and water vapour, $\delta^{18}\text{O}_{\text{IN}} = \delta^{18}\text{O}$ of imbibed water, θ = input-output flux ratio of H_2O into the body fluid; see the original article for more detailed information). To test the applicability of the model for the studied periglacial environments, we defined relative humidity during snail activity as 0.85, θ as zero, and temperature as mean of all months that correspond to a monthly average above 5 °C. The $\delta^{18}\text{O}_{\text{precipitation}}$ signals that were calculated from $\delta^{18}\text{O}_{\text{shell}}$ using this model are shown in Table 1. If compared with the equivalent $\delta^{18}\text{O}_{\text{precipitation}}$ data from the IAEA (for all months with average temperatures above 5 °C), they show a strong correlation ($\text{adjR}^2 = 0.61$, p < 0.001, see Fig. 4) with a mean rooted square error (RMSE) of 1.92‰. While the mechanistic approach provides an explanatory formula for reconstructing $\delta^{18}\text{O}_{\text{precipitation}}$, its application is constrained by the availability of input variables like temperature, which are usually not known for the (sub)fossil shells under investigation. To address this limitation, we derived an empirical regression model based on the data from our dry continental periglacial sites from Inner Asia, defined by Eqns (1a) and (1b) (see Results section).

We applied LASSO regression to test impact and relevance of additional variables beyond $\delta^{18}\text{O}_{\text{precipitation}}$ on the shell $\delta^{18}\text{O}$. The standardised coefficients indicate, that $\delta^{18}\text{O}_{\text{precipitation}}$ of the growing season has the strongest influence on $\delta^{18}\text{O}_{\text{shell}}$. The second most influential predictor in the model is growing season temperature, followed by continentality. For the applied regularisation settings, the best fitting LASSO model incorporating all selected variables explains 58% of the variance in $\delta^{18}\text{O}_{\text{shell}}$ ($\text{adjR}^2 = 0.58$). However, this represents only a marginal improvement over the model including exclusively $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ ($\text{adjR}^2 = 0.53$), suggesting that additional unconsidered factors contribute to the variability in $\delta^{18}\text{O}_{\text{shell}}$. One potential source of bias is, that all reference variables are based on modelled data. In addition to the intrinsic uncertainties associated with modelling, there is also a spatial resolution mismatch between the modelled climate data (~1 km grid cells) and the precise sample locations. Moreover, a temporal discrepancy exists between $\delta^{18}\text{O}_{\text{precipitation}}$ data from IAEA representing mean values of multiple years, and $\delta^{18}\text{O}_{\text{shell}}$ reflecting conditions from a single or only a few years, the latter being more strongly influenced by variability between specific years. This temporal mismatch likely introduces noise and limits the predictive accuracy of the model. To improve the reliability of the model, future studies should use synchronously measured $\delta^{18}\text{O}_{\text{precipitation}}$ data for calibration. Using data from the exact years of shell growth could account for year-specific variations, such as isotope depletion patterns on the leeward side of mountain ranges which depend on actual wind directions in the specific year. Such refinements could reduce standard errors and improve the

robustness of related palaeoclimate reconstructions.

5.5. Temperature estimation based on shell-derived $\delta^{18}\text{O}$ precipitation signals

$\delta^{18}\text{O}$ of precipitation is strongly correlated with temperature, which can be explained by Rayleigh distillation when air masses follow a trajectory from its vapour source to higher latitude and over continents. Rozanski et al. (1997) described an average global relationship of a decrease by $\sim 0.58\text{‰}$ in $\delta^{18}\text{O}_{\text{precipitation}}$ per 1 °C decrease in temperature. However, $\delta^{18}\text{O}_{\text{precipitation}}$ is also influenced by regional factors such as continentality, elevation and moisture source. To account for these influences when calculating temperature from $\delta^{18}\text{O}_{\text{precipitation}}$, isotope-enabled General Circulation Models (iGCMs) can be used. On the downside, iGCMs are computationally intensive and require many assumptions for different input variables. As an alternative, we provide a simplified regression-based transfer function for temperature approximation, which is based on our modern reference data for dry continental conditions:

$$T^{\circ}\text{C}_{\text{Growing season}} = (\delta^{18}\text{O}_{\text{precipitation}} > 5\text{ °C} + 15.28) / 0.24 \text{ (adj} R^2 = 0.23, \text{ SEE} = 1.15 \sim \pm 1.2\text{ °C}) \quad (4)$$

As this empirical formula is based on the modern relationship between temperature and $\delta^{18}\text{O}_{\text{precipitation}}$, it contains several assumptions and limitations. For example, the regression assumes a stable $\delta^{18}\text{O}_{\text{precipitation}}$ to temperature relationship over time, not accounting for varying climatic conditions like shifts in atmospheric circulation patterns, continentality or variations in vapour sources. Consequently, we should be aware that derived temperatures are approximations rather than high-precision reconstructions. To improve reliability, interpretations should be complemented by isotope-enabled general circulation models, which can integrate more complex spatial and temporal relationships related to $\delta^{18}\text{O}_{\text{precipitation}}$. The combination of both methods should improve the robustness of temperature reconstructions based on precipitation $\delta^{18}\text{O}$.

6. Conclusions and perspectives

In this study, we investigated the relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ in cold arid continental climates using modern reference data from Inner Asia. Our results showed a strong correlation between both variables, supporting the potential of gastropod shells as a reliable proxy for precipitation $\delta^{18}\text{O}$ in comparable settings. To facilitate $\delta^{18}\text{O}_{\text{precipitation}}$ reconstruction in continental Pleistocene loess deposits, we derived an empirical transfer function based on our modern field data. This regression model yields a rooted mean square error of 0.92‰ for the entire dataset and 0.89‰ when considering only data of the genus *Vallonia*, indicating a reasonable accuracy for applying it to (sub) fossil shells. As the regression model is based on a constant relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$, factors such as shifts in hydrological conditions, variations in snail activity phases, and taxon-related differences need to be carefully assessed when applying it in palaeoarchives. Furthermore, we propose a simplified transfer function to estimate temperatures from growing season $\delta^{18}\text{O}_{\text{precipitation}}$ based on modern calibration data. While this transfer function provides a quantitative temperature approximation, it assumes a constant relationship between $\delta^{18}\text{O}_{\text{precipitation}}$ and temperature over time, which may be biased by long-term shifts in, for example, atmospheric circulation patterns or continentality. To improve the robustness of temperature reconstructions, we therefore recommend combining this approach with isotope-enabled General Circulation Modelling (iGCM). Moreover, the underlying model for the quantitative relationship between $\delta^{18}\text{O}_{\text{shell}}$ and $\delta^{18}\text{O}_{\text{precipitation}}$ needs further improvement in terms of precision and reliability. A key issue remains the quality of the calibration data, such as the reliance on multi-year $\delta^{18}\text{O}_{\text{precipitation}}$ averages, which may not

fully capture the seasonal and interannual variability recorded in shells. Future studies should focus on using synchronously measured $\delta^{18}\text{O}$ of both shells and precipitation from the exact years of shell formation. In addition, high-resolution sclerochronological analyses could help to distinguish annual and intra-annual shell growth signals and improve the precision of palaeoclimate reconstructions. Our study demonstrates that oxygen isotopes in snail shells are a valuable proxy for reconstructing past conditions, yielding great potential for high-resolution climate reconstructions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.quascirev.2025.109440>.

Data availability

All data and/or code is contained within the submission.

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